

Untitled 5

Long Assessment 2

Iron–Carbon Phase Diagram and Phase Transformations

Total: 100 points

Number of questions: 4

Instructions

Use the same Fe–Fe₃C phase diagram for all four questions.

Assume:

- all compositions are in wt. % C;
- all alloys are cooled slowly enough to maintain equilibrium;
- the metastable iron–cementite diagram applies;
- graphite formation does not occur;
- cementite has the composition 6.70 wt. % C.

For every lever-rule calculation:

1. draw the appropriate tie line;
2. identify the composition at each end of the tie line;
3. write the lever-rule equation;
4. substitute the numerical values;
5. report the answer as a percentage.

Use the following notation:

- α : ferrite;
- γ : austenite;
- δ : delta ferrite;
- L : liquid;
- Fe₃C: cementite.

Figure 1. Iron–Iron Carbide Phase Diagram

Use one complete Fe–Fe₃C phase diagram showing:

- temperature from room temperature to the liquid region;
- composition from 0 to 6.70 wt. % C;
- all single-phase and two-phase regions;
- the eutectoid, eutectic, and peritectic points;
- the A_1 , A_3 , and A_{cm} boundaries.

Use the values read from Figure 1 unless a value is explicitly provided.

Question 1 — Interpretation of the Fe–Fe₃C Phase Diagram

25 points

A. Important compositions and temperatures

Using Figure 1, determine the approximate composition and temperature of each invariant point:

Reaction	Composition	Temperature
Peritectic		
Eutectic		
Eutectoid		

Write the corresponding reactions using phase symbols.

B. Solubility limits

Determine from Figure 1:

1. the maximum solubility of carbon in α -ferrite;
2. the maximum solubility of carbon in γ -austenite;
3. the maximum solubility of carbon in δ -ferrite;
4. the carbon composition of cementite.

Explain why carbon has a much greater solubility in austenite than in ferrite.

C. Phase identification

For each condition below, identify the equilibrium phase or phases present.

Alloy composition	Temperature
0.20 wt. % C	900°C
0.40 wt. % C	750°C
0.76 wt. % C	700°C
1.20 wt. % C	750°C
3.00 wt. % C	1200°C
4.30 wt. % C	1100°C

For every condition located in a two-phase region, draw the corresponding horizontal tie line and determine the composition of each phase.

D. Classification of alloys

Classify each composition as hypoeutectoid steel, eutectoid steel, hypereutectoid steel, hypoeutectic cast iron, eutectic cast iron, or hypereutectic cast iron:

1. 0.40 wt. % C
 2. 0.76 wt. % C
 3. 1.20 wt. % C
 4. 3.00 wt. % C
 5. 4.30 wt. % C
 6. 5.00 wt. % C
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Question 2 — Hypoeutectoid Steel

25 points

A plain-carbon steel contains

$$C_0 = 0.40 \text{ wt. \% C.}$$

The steel is initially held at 950°C and is then cooled slowly to room temperature.

A. Cooling path

Draw the vertical line corresponding to 0.40 wt. % C on Figure 1.

Identify the phases present at:

1. 950°C;
2. 800°C;
3. immediately above 727°C;
4. immediately below 727°C;
5. room temperature.

B. Development of the microstructure

Draw schematic microstructures at the following stages:

1. completely austenitic condition;
2. shortly after crossing the A_3 boundary;
3. immediately above the eutectoid temperature;
4. immediately below the eutectoid temperature;
5. room temperature.

Label:

- austenite;
- proeutectoid ferrite;
- pearlite.

Describe how the carbon concentration of the remaining austenite changes as proeutectoid ferrite forms.

C. Fractions immediately above the eutectoid temperature

Immediately above 727°C, take

$$C_{\alpha} = 0.022 \text{ wt. \% C}$$

and

$$C_{\gamma} = 0.76 \text{ wt. \% C.}$$

Calculate the mass fractions of:

1. proeutectoid ferrite;
2. austenite.

Use

$$W_{\alpha} = \frac{C_{\gamma} - C_0}{C_{\gamma} - C_{\alpha}},$$

and

$$W_{\gamma} = \frac{C_0 - C_{\alpha}}{C_{\gamma} - C_{\alpha}}.$$

D. Room-temperature microconstituents

The austenite present immediately above 727°C transforms into pearlite.

Determine the room-temperature mass fractions of:

1. proeutectoid ferrite;
2. pearlite.

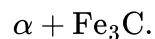
Explain why the austenite fraction immediately above the eutectoid temperature is equal to the pearlite fraction immediately below it.

E. Phases and microconstituents

Explain the difference between the following two descriptions of the room-temperature alloy:

proeutectoid ferrite + pearlite,

and



Identify which expression describes the **microconstituents** and which describes the **phases**.

Question 3 — Hypereutectoid Steel

25 points

A second plain-carbon steel contains

$$C_0 = 1.20 \text{ wt. \% C}.$$

The steel is initially completely austenitic and is then cooled slowly to room temperature.

A. Cooling path

Draw the vertical line corresponding to 1.20 wt. % C on Figure 1.

Identify the phases present:

1. above the A_{cm} boundary;

2. between the A_{cm} boundary and 727°C ;
3. immediately above 727°C ;
4. immediately below 727°C ;
5. at room temperature.

B. Development of the microstructure

Draw schematic microstructures at:

1. the completely austenitic condition;
2. shortly after crossing the A_{cm} boundary;
3. immediately above the eutectoid temperature;
4. immediately below the eutectoid temperature;
5. room temperature.

Label:

- austenite;
- proeutectoid cementite;
- pearlite.

Indicate where the proeutectoid cementite is expected to form relative to the prior-austenite grains.

C. Fractions immediately above the eutectoid temperature

Immediately above 727°C , take

$$C_{\gamma} = 0.76 \text{ wt. \% C}$$

and

$$C_{\text{Fe}_3\text{C}} = 6.70 \text{ wt. \% C.}$$

Calculate the mass fractions of:

1. austenite;
2. proeutectoid cementite.

Use

$$W_{\gamma} = \frac{C_{\text{Fe}_3\text{C}} - C_0}{C_{\text{Fe}_3\text{C}} - C_{\gamma}},$$

and

$$W_{\text{Fe}_3\text{C}} = \frac{C_0 - C_\gamma}{C_{\text{Fe}_3\text{C}} - C_\gamma}.$$

D. Room-temperature microconstituents

Determine the room-temperature mass fractions of:

1. pearlite;
2. proeutectoid cementite.

Explain why the amount of pearlite is equal to the amount of austenite that existed immediately above 727°C.

E. Comparison of the two steels

Compare the room-temperature microstructures of:

- the 0.40 wt. % C steel from Question 2;
- the 1.20 wt. % C steel.

Which alloy would generally be expected to have:

1. greater hardness;
2. greater tensile strength;
3. greater ductility?

Relate each prediction to the relative amounts of ferrite, pearlite, and cementite.

Question 4 — Eutectic Transformation and Cast Iron

25 points

An iron–carbon alloy contains

$$C_0 = 3.00 \text{ wt. \% C.}$$

The alloy is initially completely liquid and is cooled slowly to room temperature.

A. Alloy classification

Using Figure 1:

1. determine whether the alloy is a steel or a cast iron;
2. classify it as hypoeutectic, eutectic, or hypereutectic;

3. identify the primary solid phase that forms during cooling.

B. Solidification path

Draw the vertical line corresponding to 3.00 wt. % C.

Identify the phases present:

1. above the liquidus temperature;
2. between the liquidus temperature and 1147°C;
3. immediately above 1147°C;
4. immediately below 1147°C.

Describe how the compositions of the liquid and austenite change during cooling through the $L + \gamma$ region.

C. Fractions immediately above the eutectic temperature

Immediately above 1147°C, take

$$C_{\gamma} = 2.14 \text{ wt. \% C}$$

and

$$C_L = 4.30 \text{ wt. \% C.}$$

Calculate the mass fractions of:

1. primary austenite;
2. remaining liquid.

Use

$$W_{\gamma} = \frac{C_L - C_0}{C_L - C_{\gamma}},$$

and

$$W_L = \frac{C_0 - C_{\gamma}}{C_L - C_{\gamma}}.$$

D. Eutectic transformation

Write the eutectic reaction occurring at 1147°C.

Define **ledeburite** and identify the phases that constitute it.

What fraction of the alloy forms as ledeburite at the eutectic temperature?

Explain the relationship between:

- the fraction of liquid immediately above 1147°C;
- the fraction of ledeburite immediately below 1147°C.

E. Further cooling

During further slow cooling, the austenite eventually reaches the eutectoid temperature.

Write the eutectoid reaction and describe what happens to the austenite at 727°C.

Identify the equilibrium phases present at room temperature.

Explain why the phase diagram predicts the equilibrium phases and their amounts but does not directly show the shape, size, or spatial distribution of the resulting microstructure.